The 2017 tax provision primarily results from state taxes, taxes withheld in foreign jurisdictions and foreign tax expense. The 2017 tax provision differs from the statutory rate primarily due to the recognition of net operating losses as deferred tax assets, which were offset by increases in the valuation allowance, state taxes and other U.S. permanent book to tax differences.

Deferred tax assets and liabilities as of March 31, are as follows:

(In thousands)	**2019**	2018
Deferred tax assets:		
Accrued liabilities	**$ 3,944**	$ 2,720
Allowance for doubtful accounts	**120**	143
Inventory valuation reserve	**41**	20
Federal losses and credit carryforwards	**45,227**	42,713
Foreign net operating losses	**730**	623
State losses and credit carryforwards	**9,886**	9,592
Deferred revenue	**488**	652
Goodwill and other intangible assets	**—**	286
Other	**65**	96
	60,501	56,845
Less: valuation allowance	**(57,852)**	(54,260)
Total	**2,649**	2,585
Deferred tax liabilities:		
Property and equipment & software amortization	**(361)**	(412)
Goodwill and other intangible assets	**(2,706)**	(2,277)
Total	**(3,067)**	(2,689)
Total deferred tax liabilities	**$ (418)**	$ (104)

At March 31, 2019, we had $199.1 million of a federal net operating loss carryforwards that expire, if unused, in fiscal years 2031 to 2038, and $11.5 million of federal net operating loss carryforwards that can be carried forward indefinitely. Our Hong Kong, Malaysia, and Singapore subsidiaries have $0.4 million, $0.1 million, and $0.2 million of net operating loss carryforwards respectively. The losses for Hong Kong, Malaysia and Singapore can be carried forward indefinitely. At March 31, 2019 our India subsidiary had $0.4 million of minimum alternative tax credits reported as other noncurrent assets on our Consolidated Balance Sheet. Our India subsidiary operates in a “Special Economic Zone (“SEZ”)”. One of the benefits associated with the SEZ is that the India subsidiary is not subject to regular India income taxes during its first 5 years of operations. The aggregate value of the benefit of the SEZ during the current fiscal year is $0.5 million.

At March 31, 2019 we also had $127.5 million of state net operating loss carryforwards that expire, if unused, in fiscal years 2020 through 2039.

We recorded valuation allowances related to certain deferred income tax assets due to the uncertainty of the ultimate realization of the future benefits from those assets. At March 31, 2019, the total valuation allowance against deferred tax assets of $57.9 million was comprised of $57.0 million for federal and state deferred tax assets, and $0.9 million associated with deferred tax assets in Hong Kong, Malaysia, Singapore and the Philippines. In assessing the realizability of deferred tax assets, management considers whether it is more-likely-than-not that some or all of the deferred tax assets will not be realized. We have recorded a valuation allowance offsetting substantially all of our deferred tax assets. The ultimate realization of deferred tax assets depends on the generation of future taxable income during the periods in which those temporary differences are deductible. Management considers the scheduled reversal of deferred tax liabilities (including the impact of available carryback and carryforward periods), projected taxable income, and tax planning strategies in making this assessment. In order to fully realize the deferred tax assets, we will need to generate future taxable income before the expiration of the deferred tax assets governed by the tax code. Because of our losses in current and prior periods, management believes that it is more-likely-than-not that we will not realize the benefits of these deductible differences. The amount of the valuation allowance, however, could be reduced in the near term. The exact timing will be based on the level of profitability that we are able to achieve and our visibility into future results. Our recorded tax rate may increase in subsequent periods following a valuation release. Any valuation allowance release will not affect the amount of cash paid for income taxes.

The undistributed earnings of our foreign subsidiaries are not subject to U.S. federal and state income taxes unless such earnings are distributed in the form of dividends or otherwise to the extent of current and accumulated earnings and profits. The undistributed earnings of foreign subsidiaries are permanently reinvested and totaled $3.1 million and $1.7 million as of March 31, 2019 and 2018, respectively. We made the determination of permanent reinvestment on the basis of sufficient evidence that demonstrates we will invest the undistributed earnings overseas indefinitely for use in working capital, as well as foreign acquisitions and expansion. The determination of the amount of the unrecognized deferred U.S. income tax liability related to the undistributed earnings is not practicable.

Prior to the adoption of ASU 2016-09 in the first quarter of fiscal 2018, we used the with-and-without approach for ordering tax benefits derived from the share-based payment awards. Using the with-and-without approach, actual income taxes payable for the period were compared to the amount of tax payable that would have been incurred absent the deduction for employee share-based payments in excess of the amount of compensation cost recognized for financial reporting. As a result of this approach, tax net operating loss carryforwards not generated from share-based payments in excess of cost recognized for financial reporting were considered utilized before the current period's share-based deduction.

We recorded a liability for unrecognized tax positions. The aggregate changes in the balance of our gross unrecognized tax benefits were as follows for the years ended March 31:

(In thousands)	**2019**	2018	2017
Balance at April 1	**$ 687**	$ 988	$ 1,617
Reductions:			
Relating to positions taken during prior year	**—**	(300)	(604)
Relating to lapse in statute	**(107)**	(1)	(25)
Balance at March 31	**$ 580**	$ 687	$ 988

As of March 31, 2019, we had a liability of $0.6 million related to uncertain tax positions, the recognition of which would affect our effective income tax rate.

Although the timing and outcome of tax settlements are uncertain, it is reasonably possible that during the next 12 months an immaterial reduction in unrecognized tax benefits may occur as a result of the expiration of various statutes of limitations. We are routinely audited and the outcome of tax examinations could also result in a reduction in unrecognized tax benefits. Other changes could occur in the amount of gross unrecognized tax benefits during the next 12 months which cannot be estimated at this time.

We recognize interest accrued on any unrecognized tax benefits as a component of income tax expense. Penalties are recognized as a component of general and administrative expenses. We recognized interest and penalty expense of less than $0.1 million for the years ended March 31, 2019, 2018 and 2017. As of March 31, 2019 and 2018, we had approximately $0.5 million and $0.8 million of interest and penalties accrued.

In the U.S. we file consolidated federal and state income tax returns where statutes of limitations generally range from three to five years. Although we have resolved examinations with the IRS through tax year ended March 31, 2010, U.S. federal tax years are open from 2006 forward due to attribute carryforwards. The statute of limitations is open from fiscal year 2012 forward in certain state jurisdictions. We also file income tax returns in international jurisdictions where statutes of limitations generally range from three to seven years. Years beginning after 2008 are open for examination by certain foreign taxing authorities.

11. Employee Benefit Plans

401(k) Plan

We maintain 401(k) plans for employees meeting certain service requirements. Generally, the plans allow eligible employees to contribute a portion of their compensation, and we match 100% of the first 1% of the employee's pre-tax contributions and 50% of the next 5% of the employee's pre-tax contributions. We may also make discretionary contributions each year for the benefit of all eligible employees under the plans. Agilysys matching contributions were $1.6 million, $1.7 million, and $1.4 million in fiscal 2019, 2018, and 2017, respectively.

Endorsement Split-Dollar Life Insurance

Agilysys provides certain former executives with life insurance benefits through endorsement split-dollar life insurance arrangements. We entered into non-cancelable agreements with each of the former executives, whereby we must maintain the life insurance policy